

EMDEN, Germany

WMO: 102000

Lat: 53.3881N

Lon: 7.2286E

Elev: -2

StdP: 101.34

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.6	-5.6	-10.2	1.6	-6.7	-8.0	1.9	-4.8	13.2	6.7	11.7	7.0	3.3	90	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	27.6	19.7	25.3	18.9	23.3	17.9	20.6	25.7	19.6	24.0	18.7	22.2	4.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	22.3	18.0	12.9	21.6	17.2	12.3	20.5	59.3	25.8	56.0	24.1	52.9	22.3	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.6	8.5	-11.1	31.7	3.6	1.4	-13.6	32.7	-15.7	33.5	-17.7	34.3	-20.3	35.4		
(5)			-11.3	22.9	3.6	0.7	-13.9	23.4	-15.9	23.8	-18.0	24.2	-20.5	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.7	2.7	2.8	5.1	8.8	12.3	15.1	17.3	17.4	14.6	10.3	6.2	3.0
	DBStd	6.38	4.24	4.01	3.53	3.30	3.00	2.95	2.68	2.54	2.63	3.39	3.42	4.12
	HDD10.0	1029	226	202	156	60	11	1	0	0	1	38	117	218
	HDD18.3	3210	483	434	412	284	187	104	53	49	114	249	363	476
	CDD10.0	916	1	1	2	26	83	154	227	228	141	47	5	1
	CDD18.3	56	0	0	0	0	2	8	22	19	4	0	0	0
	CDH23.3	439	0	0	0	1	19	78	179	137	26	0	0	0
	CDH26.7	98	0	0	0	0	1	18	43	36	2	0	0	0
Wind	WSAvg	4.1	4.8	4.8	4.7	4.1	3.9	3.8	3.7	3.5	3.7	4.3	4.1	4.4
Precipitation	PrecAvg	785	64	43	56	46	57	72	78	73	73	67	83	69
	PrecMax	945	139	114	127	104	151	144	171	173	195	133	138	159
	PrecMin	598	2	6	10	7	11	28	10	13	14	16	21	15
	PrecStd	94	34	24	27	26	29	32	40	37	37	30	29	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.6	16.1	22.2	25.8	29.4	30.1	30.2	26.3	19.8	15.2	12.3
		MCWB	10.6	10.4	11.1	14.3	17.6	20.9	20.6	20.3	19.3	16.6	13.1	11.2
	2%	DB	10.5	10.9	13.8	19.8	22.9	25.6	27.7	26.9	23.4	17.6	13.1	10.7
		MCWB	9.5	9.3	10.3	13.2	15.9	18.5	19.7	20.0	18.2	15.1	11.8	9.7
	5%	DB	9.2	9.5	11.9	17.2	20.5	22.8	25.2	24.2	21.0	16.1	11.9	9.5
		MCWB	8.2	8.3	9.5	12.4	14.6	17.5	19.2	19.0	17.2	14.2	10.6	8.5
	10%	DB	8.2	8.2	10.3	15.0	18.3	20.5	23.0	22.1	19.0	14.9	10.7	8.4
		MCWB	7.3	7.1	8.4	11.4	13.8	16.4	18.2	17.8	15.9	13.3	9.5	7.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	11.0	12.6	15.5	19.1	21.4	22.0	22.0	20.3	16.9	13.5	11.6
		MCDB	11.6	12.4	14.9	20.1	24.0	27.4	27.1	27.4	23.9	19.4	14.7	12.0
	2%	WB	9.6	9.6	10.9	14.0	16.9	19.5	20.6	20.4	18.8	15.5	12.0	9.9
		MCDB	10.5	10.5	12.5	18.0	21.3	24.1	26.0	25.0	22.7	17.0	13.0	10.5
	5%	WB	8.4	8.4	9.8	12.8	15.4	18.1	19.6	19.2	17.5	14.5	10.7	8.6
		MCDB	9.1	9.4	11.8	16.4	19.0	21.4	24.2	22.9	20.4	15.8	11.6	9.4
	10%	WB	7.3	7.2	8.5	11.6	14.2	16.9	18.5	18.4	16.4	13.6	9.7	7.6
		MCDB	8.0	8.1	10.1	14.8	17.5	20.0	22.1	21.5	18.5	14.8	10.6	8.3
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.8	6.5	8.7	8.6	8.2	7.8	7.7	7.4	6.2	4.5	4.1
		MCDBR	4.3	5.4	9.4	13.1	13.3	12.7	13.0	11.8	11.0	7.1	5.0	4.3
		MCWBR	4.0	4.4	6.6	7.7	7.3	6.7	6.2	5.9	6.4	5.1	4.2	4.2
	5% WB	MCDBR	4.3	4.7	8.6	11.6	11.4	10.7	11.5	10.3	10.2	6.9	4.9	4.2
		MCWBR	4.1	4.1	6.4	7.3	7.1	6.3	6.0	5.6	6.3	5.3	4.4	4.2
Clear-Sky Solar Irradiance	taub	0.328	0.347	0.375	0.407	0.404	0.403	0.415	0.410	0.388	0.362	0.342	0.320	
	taud	2.317	2.331	2.272	2.237	2.284	2.311	2.290	2.318	2.373	2.404	2.350	2.309	
	Ebn at Noon	633	745	799	821	845	849	828	810	781	719	609	563	
	Edh at Noon	61	81	105	123	123	122	122	113	95	75	59	51	
All-Sky Solar Radiation	RadAvg	0.59	1.24	2.46	4.10	5.09	5.35	5.07	4.21	2.99	1.67	0.72	0.45	
	RadStd	0.06	0.19	0.26	0.40	0.42	0.39	0.59	0.34	0.30	0.17	0.07	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-122	N/A	N/A

Nomenclature: See separate page